

in sight. SixthSense is a remarkable device; it uses these off the shelf components and costs approximately \$350 to build (not including the computer). The micro-projector and the camera are connected to a mobile computing device placed in the user's pocket. The mini-projector projects visual images onto your surroundings- walls, hand, table, etc., enabling them to be used as interactive screens. The camera and mirror are used to recognize and track a user's hand gestures, colors and physical objects using image-processing based techniques. The software on the computing device processes the video stream data captured from the camera using simple computer-vision techniques, gathers GPS coordinates and pulls up data from the Internet. The user may also wear colored caps on the fingers which are used to interact with the projected image; even fingers with different colors of nail polish would work. The movements and arrangement of these colored markers (also known as fiducials) and hand gestures serve as interaction instructions for the projected application interfaces. SixthSense also supports multi-touch interaction letting you zoom in, zoom out or pan using many of the intuitive hand movements that we are now familiar with. The SixthSense prototype implements a number of demonstrative applications:

- ▶ The map application lets you navigate a map displayed on a nearby surface using hand gestures.
- ▶ The drawing application lets you draw on any surface by tracking the fingertip movements of the user's index fingers.
- ▶ SixthSense also lets you draw symbols in the air using the tip of your index finger and recognizes those symbols as interaction instructions. Drawing a magnifying glass symbol takes you to the map application or drawing an '@' symbol lets you check mail; Draw a circle on your wrist and SixthSense projects an analog watch on it.
- ▶ SixthSense recognizes your complex hand gestures. The SixthSense system takes photos of the scene you are looking at by detecting the 'framing' gesture. You can stop by any surface or wall and flick through the photos you've taken.
- ▶ The SixthSense system also augments physical objects in your vicinity by overlaying more information on them and making them interactive. For example, a newspaper can show live video news or dynamic information provided on a regular piece of paper. Touch any object and learn so much more about it than appears to the eye.

For more details on SixthSense log on to: <http://dgit.in/Xn9EZw>.